Unified Agentic Environments and Task Synthesis

Deep Dive into Kimi K3 — Chapter 24

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Foundation Books Project

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Goal: type tasks, environments, harnesses, and traces; turn concepts into executable, independently checked tasks; follow persistent and investigative episodes into policy learning — and keep each claim in its evidence class.

Roadmap

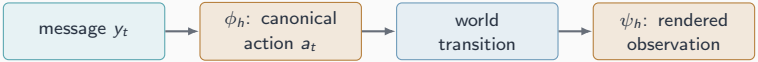
Harnesses and typed environments

From concepts to executable tasks

Persistence, investigation, and learning

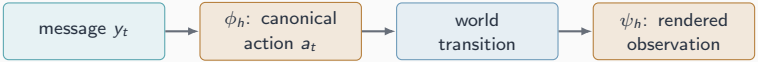
Harnesses and typed environments

The harness changes the decision problem



- The same inspection, spelled `inspect_boxes` or `assets.inspect_bboxes`, is a different bound task $b_i = (x_i, \mathcal{E}_{\nu_E}, h, \nu_b)$. [REPORTED: K3 Sec. 4.2.1]

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- Adapter failure, illegal action, and failed evaluation are three boundaries; family and harness marginals can balance while joint cells stay empty.

From concepts to executable tasks

Concepts, materials, and admitted tasks

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- Admission needs an initial world, goal, learner-visible feasibility witness, and independent completion test; pooled and hard-coded repairs fail 6/10. [OBSERVED: companion checks]

Persistence, investigation, and learning

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Construction validity, interface equivalence, and learned transfer are separate claims. Next: Chapter 25's Visual and Multimodal Agent Learning.